SRSM Optimization of a Plate Geometry

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Objective and design space. The problem is to maximize the absorbed deformation energy E_{abs} of the plate, as computed by the FreeCAD FEM workflow, while satisfying geometric and stress feasibility constraints. The design vector is

$$\mathbf{x} = (x_1, y_1, x_2, y_2, x_3, y_3, \theta),$$

where the first six variables are cutout-center coordinates in mm and θ is the capsule angle in degrees. The parameter bounds were chosen as a local, engineering-feasible search window around promising preliminary designs while respecting the CAD clearance limits. This keeps the expensive FEM budget focused on realistic geometries, effectively decreases the number of geometry-infeasible samples and improves surrogate accuracy, while still leaving enough range for SRSM to adapt the cutout positions and capsule angle. The chosen bounds are shown in the following table:

Variable	Meaning	Bound
x_1	circle 1 x	$[39.0, 45.0]$
y_1	circle 1 y	$[45.0, 50.5]$
x_2	circle 2 x	$[40.0, 47.0]$
y_2	circle 2 y	$[22.0, 28.5]$
x_3	capsule x	$[17.0, 23.0]$
y_3	capsule y	$[27.5, 33.0]$
θ	capsule angle	$[62.0, 73.0]$

Constraints and feasibility. Geometry is screened before FEM: minimum edge clearance is 2 mm and minimum cutout-to-cutout clearance is 1 mm. Stress feasibility is checked after FEM using

$$\sigma_{\max} \leq 275 \text{ MPa}.$$

This prefilter avoids wasting expensive solver calls on impossible geometries; stress-infeasible FEM rows are retained in the log but are not allowed to update the best feasible design.

Optimization strategy. A budgeted successive response surface method (SRSM) was used because each FEM call is expensive. The run began with a 70-point geometry-valid Latin-hypercube DOE generated from 90 raw candidates. FEM-confirmed DOE and SRSM results were then used to fit adaptive response surfaces: quadratic when enough local data were available and linear otherwise. In each SRSM iteration, 500 geometry-valid candidates were sampled in the current region of interest (ROI), energy and stress were predicted cheaply, a stress-penalized acquisition score ranked the candidates, and the top 3 candidates were evaluated by FreeCAD/FEM. Surrogate predictions were used only for ranking; the reported optimum is based on confirmed FEM evaluations.

ROI adaptation. The active ROI was centered on the current best FEM-feasible design and clipped to the global bounds. Its side-length fraction started at 1.00, shrank by 0.95 after improvement and by 0.90 after no improvement, and was bounded below by 0.25. This gives early exploration over the full domain and later exploitation near the best confirmed design.

Numerical result. The run used 148 FEM calls out of a budget of 160: 70 DOE calls and 78 surrogate-ranked SRSM calls over 26 SRSM iterations. All 148 logged FEM calls were geometry-valid; 135 were fully feasible and 13 exceeded the stress limit. The best DOE objective was 13015.487, while the final FEM-confirmed best objective was

$$E^*_{\text{abs}} = 13129.789$$

at evaluation 129, SRSM iteration 20. The corresponding stress was $\sigma_{\max} = 230.919$ MPa, safely below the 275 MPa limit. The final best design was

$$\mathbf{x}^* = (42.373,\ 45.038,\ 44.814,\ 28.353,\ 19.669,\ 32.823,\ 72.953).$$

Relative to the best DOE result, the confirmed best-so-far improved by 114.302 energy units, or 0.878%. The run stopped when the ROI reached the minimum fraction 0.25 and no meaningful improvement occurred for 6 consecutive SRSM iterations.

Plot interpretation. The convergence plot shows early surrogate optimism, including a predicted peak of about 13288.65 at iteration 3 that was not FEM-confirmed. The FEM-evaluated curve fluctuates because each iteration tests new ranked candidates in a nonlinear constrained black-box problem; it should not be read as a monotone best-so-far curve. Scientifically, the important behavior is that the confirmed best improves from the DOE value to 13129.789 and then plateaus: iterations 21–26 give per-iteration best FEM values between 13116.94 and 13127.56, below the iteration-20 best.

The ROI plot shows the search localizing around high y_2, high y_3, and high angle values near the upper global angle bound. The final ROI fraction is 0.25, with final angle bounds $[70.25, 73.00]^\circ$. This supports local stabilization of the SRSM search, but it does not prove global optimality.

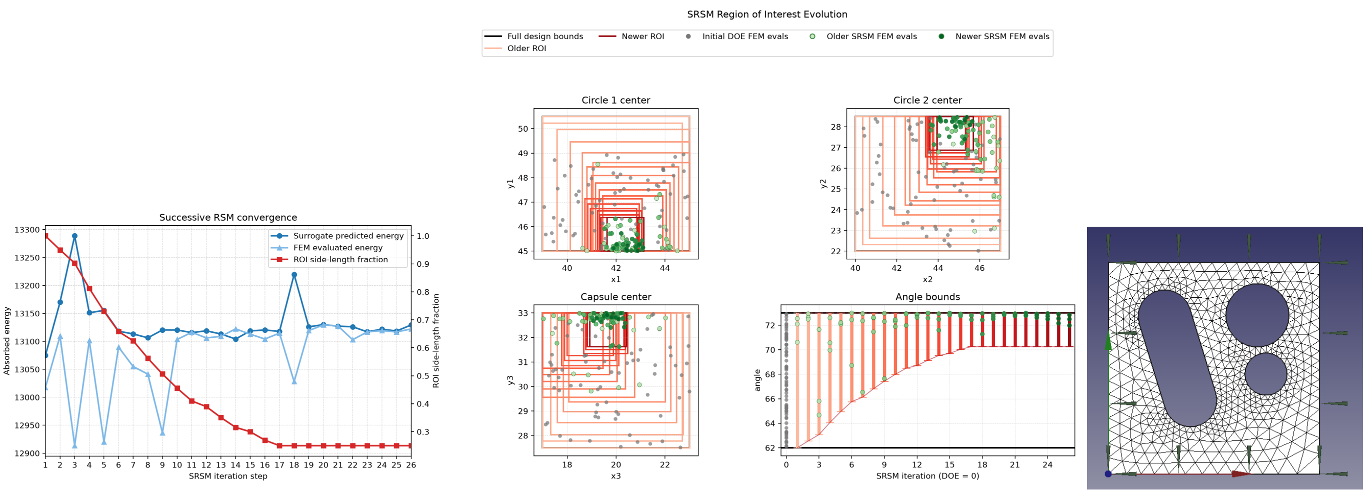

Figure 1: Left: SRSM convergence history; surrogate predictions rank candidates, while FEM values are confirmed black-box evaluations. **Middle:** region-of-interest (ROI) contraction and recentering in the design space. **Right:** FreeCAD view of the final FEM-confirmed optimum.